WORKED SOLUTION

STEP I 1994, Question 14

Probability and Statistics • Exponential Order Statistics

STEP I 1994, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s1-q14

The key idea

Each T_i is exponentially distributed, so part (i) is a pair of standard integrals by parts. For the last and first essays, the trick is to turn a statement about a maximum or a minimum into a statement about *every* T_i: the last essay is in by time u exactly when all of them are, and the first one is still outstanding at time t exactly when all of them are. Independence then turns "all" into a product, and differentiating the distribution function gives the density.

Solution

Part (i)

Write X for any one of the T_i, with density $f(t) = \lambda \mathrm{e}^{-\lambda t}$ for $t \geq 0$. Integrating by parts, with $u = t$ and $\mathrm{d}v = \lambda \mathrm{e}^{-\lambda t}\,\mathrm{d}t$ (so $v = -\mathrm{e}^{-\lambda t}$),

$$\mathrm{E}(X) = \int_0^\infty t\,\lambda \mathrm{e}^{-\lambda t}\,\mathrm{d}t = \Big[-t\mathrm{e}^{-\lambda t}\Big]_0^\infty + \int_0^\infty \mathrm{e}^{-\lambda t}\,\mathrm{d}t = 0 + \frac{1}{\lambda} = \frac{1}{\lambda},$$

since $t\mathrm{e}^{-\lambda t} \to 0$ as $t \to \infty$ (exponentials beat powers). In the same way,

$$\mathrm{E}(X^2) = \int_0^\infty t^2 \lambda \mathrm{e}^{-\lambda t}\,\mathrm{d}t = \Big[-t^2\mathrm{e}^{-\lambda t}\Big]_0^\infty + \int_0^\infty 2t\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t = \frac{2}{\lambda}\,\mathrm{E}(X) = \frac{2}{\lambda^2},$$

where we recognised $\int_0^\infty t\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t$ as $\frac{1}{\lambda}\mathrm{E}(X)$. Therefore

$$\mathrm{E}(T_i) = \frac{1}{\lambda}, \qquad \mathrm{Var}(T_i) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

Part (ii)

First we need the distribution function of a single T_i: for $u \geq 0$,

$$\mathrm{P}(T_i \leq u) = \int_0^u \lambda \mathrm{e}^{-\lambda t}\,\mathrm{d}t = 1 - \mathrm{e}^{-\lambda u}.$$

Now $U = \max(T_1, \ldots, T_n)$, and the largest of the T_i is at most u if and only if every one of them is at most u:

$$\{U \le u\} = \{T_1 \le u\} \cap \{T_2 \le u\} \cap \cdots \cap \{T_n \le u\}.$$

Ask yourself Why is it legitimate to multiply the probabilities here?

The events $\{T_i \le u\}$ concern different T_i, which are independent by assumption, so the probability of the intersection is the product of the probabilities. Hence, for $u \ge 0$,

$$\mathrm{P}(U \le u) = \prod_{i=1}^{n} \mathrm{P}(T_i \le u) = (1 - \mathrm{e}^{-\lambda u})^n,$$

as required. The density of U is the derivative of this distribution function; by the chain rule,

$$f_U(u) = n(1 - \mathrm{e}^{-\lambda u})^{n-1} \cdot \lambda \mathrm{e}^{-\lambda u} = n\lambda\, \mathrm{e}^{-\lambda u}(1 - \mathrm{e}^{-\lambda u})^{n-1} \qquad (u \ge 0),$$

and $f_U(u) = 0$ for $u < 0$.

Part (iii)

Now $T = \min(T_1, \ldots, T_n)$. Here the useful event is the complement: the first essay arrives after time t if and only if *none* has arrived by then, that is, every $T_i > t$. With $\mathrm{P}(T_i > t) = \mathrm{e}^{-\lambda t}$ and independence again,

$$\mathrm{P}(T > t) = \prod_{i=1}^{n} \mathrm{P}(T_i > t) = (\mathrm{e}^{-\lambda t})^n = \mathrm{e}^{-n\lambda t} \qquad (t \ge 0).$$

So $\mathrm{P}(T \le t) = 1 - \mathrm{e}^{-n\lambda t}$, and differentiating,

$$f_T(t) = n\lambda\, \mathrm{e}^{-n\lambda t} \qquad (t \ge 0),$$

with $f_T(t) = 0$ for $t < 0$.

Part (iv)

The density in (iii) is exactly the density of part (i) with λ replaced by $n\lambda$: T is exponential with parameter $n\lambda$. So we can simply write down, from part (i),

$$\mathrm{E}(T) = \frac{1}{n\lambda}, \qquad \mathrm{Var}(T) = \frac{1}{n^2\lambda^2}.$$

This makes sense: with n students, the first essay should arrive about n times sooner than any particular one.

Checks

- With $n = 1$ there is only one essay, so $T = U = T_1$; indeed both densities reduce to $\lambda \mathrm{e}^{-\lambda t}$.

- $\mathrm{P}(U \leq u)$ rises from 0 at $u = 0$ to 1 as $u \to \infty$, as a distribution function must.
- We simulated 2×10^5 classes with $n = 5$, $\lambda = 0.7$. For example, the proportion with $U \leq 2$ was 0.2425 against $(1 - \mathrm{e}^{-1.4})^5 \approx 0.2427$, and the mean of T was 0.2860 against $1/(5 \times 0.7) \approx 0.2857$. Simpson's rule confirmed that f_U integrates to 1.

Common slips

- Working with $\mathrm{P}(T \leq t)$ directly for the minimum: "at least one $T_i \leq t$" is not an intersection, so it does not factorise. Go via $\mathrm{P}(T > t)$.
- Forgetting to state that independence is what justifies the product.
- Writing $\mathrm{Var}(T_i) = \mathrm{E}(T_i^2) = 2/\lambda^2$ and forgetting to subtract $(\mathrm{E}\,T_i)^2$.

Beyond STEP

T and U are the smallest and largest *order statistics* of the sample. The minimum of independent exponentials being exponential with the rates added is the basis of Poisson processes and queueing theory. A lovely further fact, from the memoryless property, is that the gaps between successive essays are independent exponentials with rates $n\lambda, (n-1)\lambda, \ldots, \lambda$, so $\mathrm{E}(U) = \frac{1}{\lambda}\left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right)$. Both of these are developed in Grimmett and Welsh, *Probability: An Introduction*, and in Blitzstein and Hwang, *Introduction to Probability*.

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